AWS Cost Optimization using Lambda & EventBridge

🚀 Project Overview

This project demonstrates for reducing AWS EC2 costs by automating instance start and stop operations using AWS Lambda and Amazon EventBridge. The solution eliminates unnecessary EC2 runtime during non-business hours through scheduled, serverless automation.

🎯 Problem Statement

Running EC2 instances 24/7 leads to unnecessary compute costs, especially when workloads are only required during specific hours. Manual start and stop operations are inefficient and error-prone.

🏗️ Workflow

EventBridge (Cron Schedule)

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AWS Lambda (Start/Stop Logic)

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Amazon EC2

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CloudWatch Logs & CloudTrail (Verification)

⚙️ Implementation Highlights

Automated EC2 lifecycle management using **AWS Lambda**

Scheduled execution using **Amazon EventBridge (Cron)**

Secure, keyless access via **IAM roles**

Execution monitoring with **CloudWatch Logs**

Audit-level verification using **AWS CloudTrail**

🛠️ AWS Services and Technology Used

AWS Lambda

Amazon EventBridge

Amazon EC2

AWS IAM

Amazon CloudWatch Logs

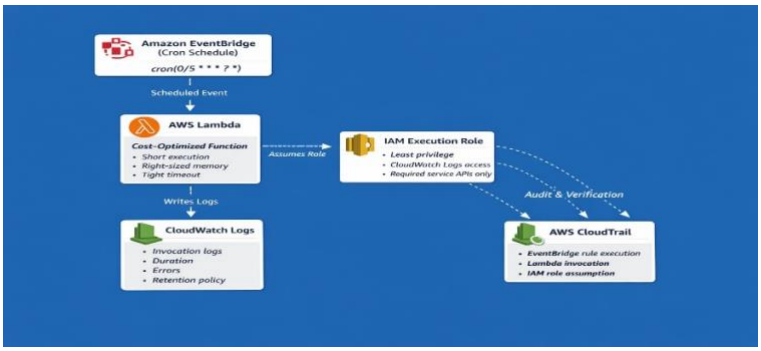
AWS CloudTrail

Python(Boto3)

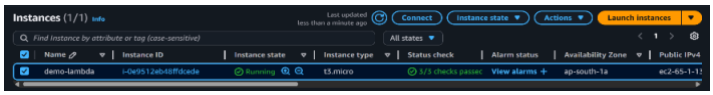
💡 Key Outcome

- Reduced unnecessary EC2 runtime
- Improved cost efficiency through automation
- Fully serverless, scalable, and production-ready design

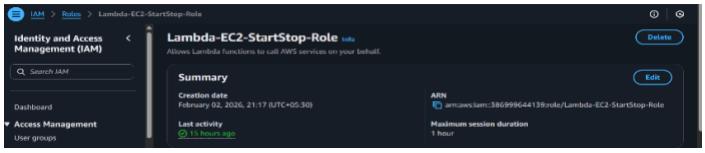
📊 Architecture flow

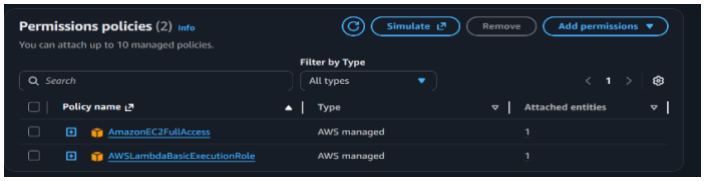


Screenshots: Launch an EC2 instance:

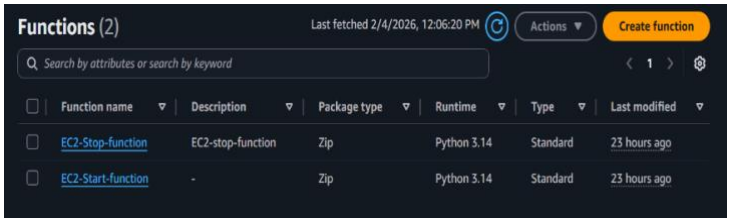


Create and attach IAM role and policies

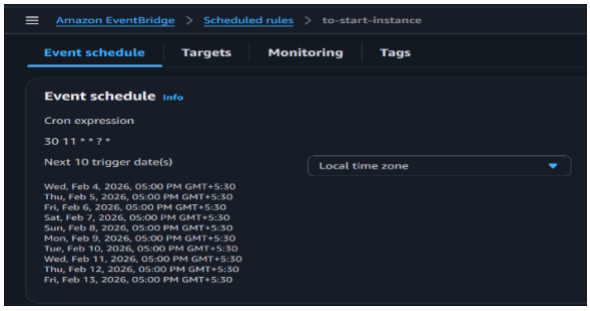




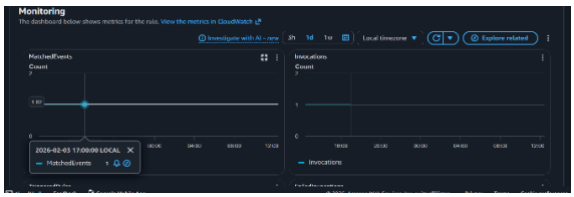
Create lambda functions:



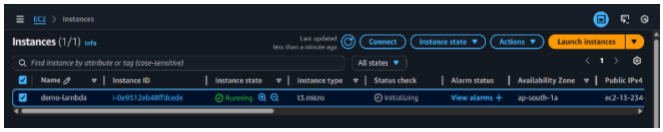
Event Schedule to **START** the instance:



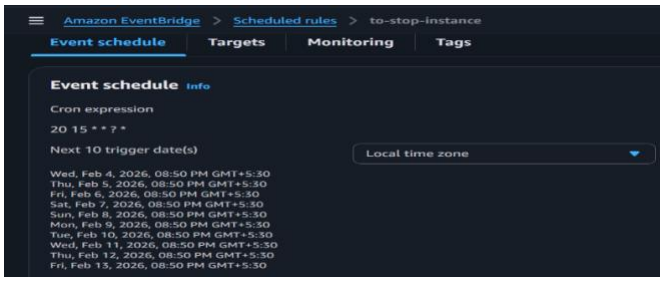
CloudWatch Monitoring:



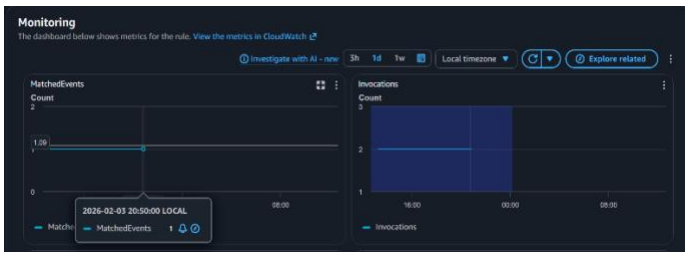
Instance state: Initialization of status check pass



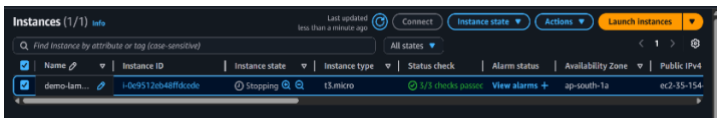
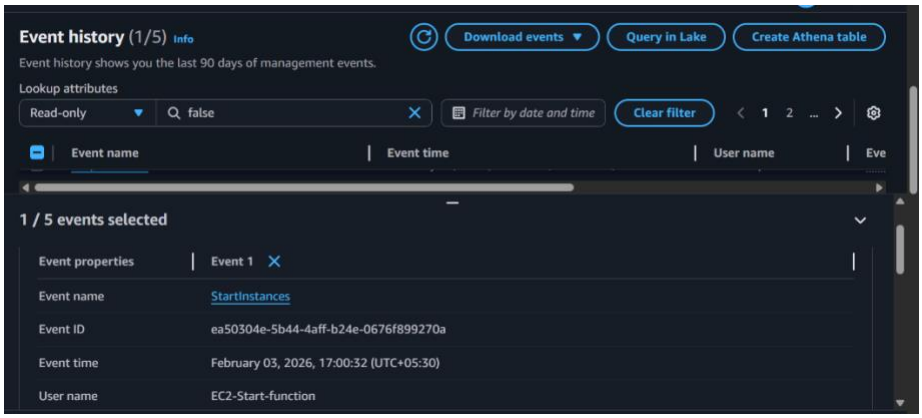
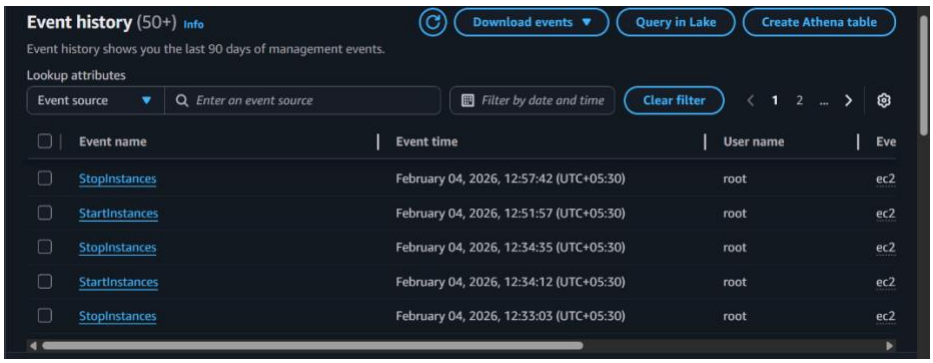
Similarly, Event Schedule to **STOP** the instance:



CloudWatch Monitoring:



Instance State: Instance state is getting stopped



CloudTrail Logs: